

MATCC: A Novel Approach for Robust Stock Price Prediction Incorporating Market Trends and Cross-time Correlations

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Abstract

Stock price prediction has been a challenging problem due to non-stationary dynamics and complex market dependencies. Existing work has two limitations: 1. Previous studies have underestimated the importance of market trends, relying solely on stock data to learn patterns and capture market regularities implicitly. However, due to random stock fluctuations and trading noise caused by market sentiment, it is difficult to learn underlying market trends, resulting in poor model performance. 2. Prior research has predominantly concentrated on time-aligned feature correlations, with limited exploration of cross-time stock correlations. To address these issues, we propose a novel framework, MATCC (Market Trend and Cross-time Correlation model). It explicitly extracts market trends as guiding information, decomposes stock data into trend and fluctuation components, and employs a carefully designed structure for mining cross-time correlation. Extensive experiments demonstrate that MATCC significantly outperforms previous works in both ranking and portfolio-based metrics. Additionally, we illustrate the influence of trends and correlations on stock prediction through visualization. We publish our code at <https://github.com/caozhiy/MATCC>.

CCS Concepts

• **Computing methodologies** → **Machine learning; Artificial intelligence**; • **Applied computing** → *Economics*.

Keywords

computational finance; stock prediction; volatility forecasting

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1 Introduction

Stock price forecasting is a crucial technique for profitable stock investment by leveraging historical data from the stock market to predict future trends. This practice plays a pivotal role in guiding investment decisions and maximizing returns. Accurate stock price forecasting helps investors to make informed investment decisions, allows financial institutions to price derivatives, and lets regulators manage the amount of risk in the financial system. As a result, the task of stock market prediction has become increasingly challenging and essential in finance, which has attracted significant attention from academia and industry [9, 15, 18, 39].

In the field of stock prediction, previous work [9, 30, 32, 33, 37] has relied on predefined concepts, relationships, or rules and created a static correlation graph. While these methods provide insight into the relationships between stocks, they also greatly limit the types of relationships between stocks and do not explain real-time correlations between stocks. Furthermore, pre-defined relationships may not generalize to new stocks in an evolving market. The Transformer architecture [28] uses a self-attention mechanism to compute dynamic stock correlations, which is more flexible and applicable to the set of stocks in the market that change over time. However, these models lack effectiveness in capturing both trends and correlations.

In stock price prediction, market and stock trends are critical features that help models better understand the historical relationship between stock prices and broader market movements. However, many existing methods downplay the importance of market and stock trends. They tend to emphasise individual stock properties [7] such as closing price and trading volume from stock data only, and market trends are often implicitly captured by aggregating all individual stock features [22, 35, 41]. However, stock data often contains significant noise [10], i.e. short-term random fluctuations or non-systematic components of price movements. Such fluctuations can be caused by factors such as market sentiment, unforeseen events, and uncertainty in trading activity rather than changes in fundamentals or long-term trends [2]. This noise can distort patterns in

the data, making it difficult for the model to identify underlying market trends and trading signals, leading to poor forecasts.

In addition, existing models focus more on time-aligned stock correlations across all input features, but stock correlations often occur in an inter-temporal manner [16]. For example, the typical approach usually learns temporal embeddings from individual stock time series and then blends temporal representations (time-aligned) to construct stock embeddings for future modelling stock correlation. This method tends to blur the detailed features of stocks at each point in time and over-simplifies the unique properties of individual stocks at different time steps. Specifically, stock correlations are highly dynamic and may exist at different time steps [1]. This direct aggregation of temporal representations reduces the model's ability to capture stock correlations.

To address these challenges, we introduce a novel framework, MATCC (Market Trends and Cross-time Correlation model). It explicitly extracts the market trend as the global environmental information through a convolution operation and allows each stock to receive guidance from the market trend. At the same time, we decompose the stock data into trend and fluctuation components to re-emphasize the stock trend and separate the noisy volatility component. We carefully design the correlation structure to extract stock cross-time correlations. By maintaining a sequence of temporal representations for each stock, we preserve each time step's unique temporal attributes and characteristics, enabling correlation modelling in the time and stock dimensions.

The main contributions of this paper are summarized as follows:

- We propose a novel market-stock trend guidance module to extract effective market movement features and capture long-term trends as well as fluctuations in stock time series data, thereby effectively reducing the volatility of model results and improving stock forecasting capabilities.
- We design a sophisticated correlation module that maintains a sequence of temporal representations for each stock to capture the cross-time correlation and better identify the linkages between stocks.
- We have conducted rich experiments to validate the designs of our proposed method MATCC and demonstrated its superiority over baselines. Through visualization, we explore the impact of trends and correlations on stock prediction.

2 Related Works

2.1 Stock Correlation Model

Stock correlation modelling [41] has been an integral research area in stock price prediction. Many studies have shown that deep-learning-based stock correlation models outperform traditional machine learning. Researchers have started to consider stock feature sequences and learn time correlation automatically. They introduced various time series models, including RNN-based models [14, 43], CNN-based models [30, 31], and attention-based models [4, 19] to reveal the internal time dynamics of stocks. Recent research focuses on modelling inter-stock correlations and constructing dynamic implicit stock relationships for highly volatile markets, including the use of graph-based [22], hypergraph-based [14, 24], and attention-based modules [36, 41] to fuse temporal and spatial features of stocks. In particular, MASTER [16] models both instantaneous and

inter-temporal stock correlations by alternating intra- and inter-stock aggregation and introduces market information to screen for valid features automatically.

2.2 Time Series Decomposition

Time series decomposition [13] is a technique to decompose time series data into trend, seasonality and residual components. This technique helps understand the long-term trends and cyclical and stochastic fluctuations in time series data for better forecasting and analysis. Commonly used methods include Classical Decomposition, Seasonal and Trend decomposition using Loess [26]. e.g., AutoFormer [34] gradually extracts the long-term trend from the predicted hidden variables using sequence decomposition blocks. FEDformer [44] is further improved by applying decomposition algorithms in frequency rather than time by applying the decomposition algorithm.

3 Methodology

In this section, we first formulate the task of the stock price prediction. Then, the proposed MATCC model is presented, illustrated in Figure 1. There are four main components in the framework: (1) **Market-Stock Trend Guidance Module**. The Market-Stock Trend Module extracts market information for individual stock guidance. We use time series decomposition to capture each stock's trend and fluctuation components in the lookback window, smoothing out potential noise. (2) **Correlation Module**. In the Correlation Module, we maintain a series of temporal representations of each stock and model cross-time correlations from both time and stock dimensions. (3) **Stock Embedding Aggregation Module**. We compute the final representation of the stock by combining the last time embedding with the historical time embedding. (4) **Prediction**. The comprehensive stock embedding is sent to prediction layers for label prediction. In the following subsections, we elaborate on the details of MATCC step by step.

3.1 Problem Formulation

Based on previous work in stock analysis [8, 14, 25], we adopt the problem definition given by [16]. Given a set of stocks $\mathcal{S}$, features of each stock u on each trading day $t \in [1, T]$ are collected to form a feature vector $x_{u,t} \in \mathbb{R}^F$. We focus on predicting the stock return ratio because it normalizes the market price variety between different stocks compared to the absolute price change. The return ratio is $\hat{r}_u = (c_{u,T+d} - c_{u,T+1}) / c_{u,T+1}$, where $c_{u,t}$ is the closing price of stock u at time step t , and d represents the predetermined prediction interval. We perform daily Z-score normalization of the return ratio as in previous work [40], $r_u = \text{Norm}_{\mathcal{S}}(\hat{r}_u)$. The stock forecasting task can be formally described as: Given stock features $\{x_{u,t}\}_{u \in \mathcal{S}, t \in [1, T]}$, the stock price forecasting is to jointly predict the future normalized return ratio $\{r_u\}_{u \in \mathcal{S}}$.

3.2 Market-Stock Trend Guidance Module

3.2.1 Market Status Representation. Firstly, we construct a market status representation vector to describe the current trading situation of the market. In contrast to [41], which collects factors of individual stocks to construct context about the market environment, we use indexes from exchanges and financial institutions'

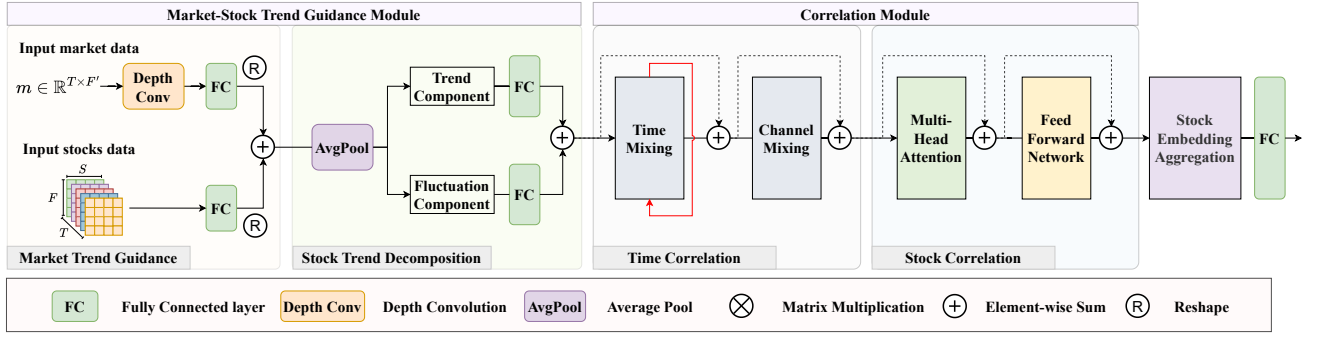


Figure 1: The overview of MATCC framework. S, T, F, F' represent stock number, lookback window length, stock features and market features respectively.

releases as market state features. Following [16], we adopt **market index price** and **market index trading volume** as the representative features. Typically, the market index price denotes a composite index specific to a market, representing the weighted mean price of a selection of stocks. Its purpose is to gauge the collective performance of those stocks within that particular market. Market index trading volume signifies the cumulative quantity of a specific market index traded during a specified time step. This metric reflects the extent of engagement among market participants in the purchasing and selling of constituent stocks within the index. Both the present and past market index prices are incorporated, with the historical data characterized by the mean and standard deviation of the preceding d' days. This approach illuminates the volatility of prices. The parameter d' defines the duration of the interval considered for incorporating historical market data into the analysis. Additionally, we include the mean and standard deviation of the market index volume over the past d' days to elucidate the market's actual scale.

3.2.2 Market Trend Guidance. After constructing the daily market status representations, we explicitly extract the market trends as the global context. In contrast to [7], which uses pooling operations to merely aggregate features, we employ one-dimensional depth convolution [38] to capture market trend information. One-dimensional depth convolution learns different feature mappings through convolution kernels, each of which can independently capture a channel's pattern. This operation avoids conflating channel trend information and preserves each channel's independence. We use a convolution kernel with $\text{kernel} = \text{stride} = k_c, \text{padding} = 0$ to perform a convolution operation, which is then linearly mapped to obtain $mt \in \mathbb{R}^{D \times T}$. The process can be formulated as:

$$m' = (W_\alpha m + b_\alpha)^T, \quad mt = W_\beta \text{DepthConv}(m') + b_\beta, \quad (1)$$

where $m \in \mathbb{R}^{T \times F'}$ denotes the market status embedding, $mt \in \mathbb{R}^{D \times T}$ denotes the market trend features, $W_\alpha, b_\alpha, W_\beta, b_\beta$ are trainable parameters and DepthConv represents Depth Convolution. F' and D represent market features and model feature dimension.

Unlike previous studies [16] that used gating mechanism to generate scaling coefficients to find the most significant feature changes, we emphasise the influence and guidance of the market trends on all features of individual stocks. Specifically, stock returns in the

market are stimulated or dampened by the overall trading market, and each stock's representation should explicitly include these characteristics. In practice, we directly sum mt and individual stock x_u to obtain the market-trend-guided stock feature $\hat{x}_u \in \mathbb{R}^{T \times D}$, $\hat{x}_u = W_\theta x_u + mt^T$, where W_θ is a trainable parameters.

3.2.3 Stock Trend Decomposition. Within the Stock Trend Decomposition module, we explicitly extract the Trend component and Fluctuation component of the stock's features via time series decomposition. The Trend component embodies stability and exhibits a long-term trajectory, such as the trend of returns, while the less stable Fluctuation component reflects short-term volatility. We decompose the input using AvgPool with kernel size k_a , which blurs local irregularities and makes the output features smoother, thus reducing the sensitivity to noise. The process can be formulated as:

$$\hat{x}_u^t = \text{AvgPool}(\text{Padding}(\hat{x}_u)), \quad \hat{x}_u^f = \hat{x}_u - \hat{x}_u^t, \quad (2)$$

where $\hat{x}_u^t, \hat{x}_u^f \in \mathbb{R}^{T \times D}$ represent the long-term trend and short-term fluctuation of the stock. Padding is used to keep the series length unchanged. Similar to [42], two one-layer linear layers are applied to each component along the time axis to further increase the expressiveness of the trend feature, and we sum the two features to produce the final result $X_u \in \mathbb{R}^{T \times D}$. It can be represented as:

$$X_u^T = \left(W_f (\hat{x}_u^f)^T + b_f \right) + \left(W_t (\hat{x}_u^t)^T + b_t \right), \quad (3)$$

where W_f, b_f, W_t, b_t are trainable parameters in linear layers, and D denotes model feature dimension.

Through market trend guidance and stock trend decomposition, the stock embedding includes long-term trends and short-term fluctuations, with market environment guidance and retained local detail, which ensures a comprehensive representation conducive to modelling stock correlations.

3.3 Correlation Module

3.3.1 Time Correlation. We first model cross-time correlation under the same stock. Specifically, we interact the information between different time steps t within each stock $X_u, u \in S$. In contrast to previous work [16] that used the self-attention method to compute the attention weights of each time step for all other time

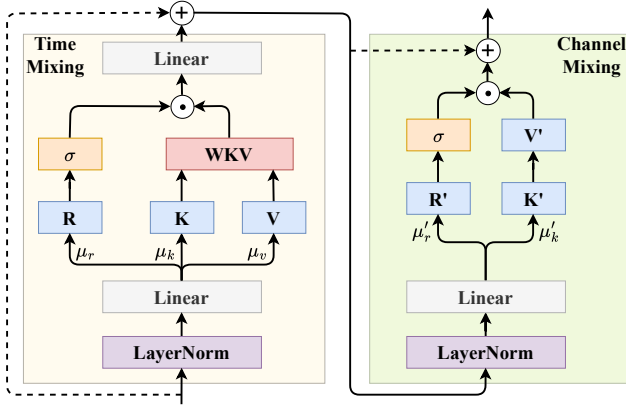


Figure 2: The RWKV structure.

steps within the lookback window, we use an RNN-based attention mechanism, RWKV [21] instead of Transformer [28] to model inter-temporal correlations. RWKV has three key advantages: (1) By employing an RNN structure, the framework maintains the temporal order of input stock features, eliminating the need to use positional encoding to compensate for temporal order within the backtracking window. (2) The causal attention mechanism [6] ensures that at any time step, the model only considers the current and previous time steps, preventing future information from influencing past data. This mechanism aligns with the inherent temporal nature of stock data, ensuring that earlier events influence later events rather than the reverse. (3) The internal attention computation employs a unique score-updating process, including a time-dependent softmax, with linear time and space complexity.¹

As shown in Figure 2, the RWKV backbone contains a time-mixing sub-block and a channel-mixing sub-block. The input of each sub-block is the time-step features of the current and previous moments, which is achieved by token shift, a simple and intuitive way of time mixing.

In the Time Mixing sub-block, we combine the current input with the previous input using the trainable parameters μ_r, μ_k, μ_v . The result is linear projected to get the receptance vector r , the keyword vector k , and the value vector v :

$$\begin{aligned} r_t &= W_r \cdot (\mu_r \odot X_{u,t} + (1 - \mu_r) \odot X_{u,t-1}), \\ k_t &= W_k \cdot (\mu_k \odot X_{u,t} + (1 - \mu_k) \odot X_{u,t-1}), \\ v_t &= W_v \cdot (\mu_v \odot X_{u,t} + (1 - \mu_v) \odot X_{u,t-1}). \end{aligned} \quad (4)$$

In RWKV, the WKV operator mirrors to self-attention but with a linear time and space complexity. It includes a time-dependent attention score update process formulated as:

$$wkv_t = \frac{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} \odot v_i + e^{u+k_t} \odot v_t}{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} + e^{u+k_t}}. \quad (5)$$

Here, w and u are two trainable parameters, $t \in [1, T]$ is the current time stamp index, $\odot$ represents element-wise multiplication. u acts as a bonus, encouraging the model to prioritize attention on

encountering a token for the first time. w represents a channel-wise time decay vector per head, managing attention decay over time across different channels. WKV also supports multi-head attention operations, which is $wkv_t = \text{concat}\{wkv_t^1, \dots, wkv_t^N\}$, where N denotes the number of heads.

The output vector o_t post the WKV operator is given by:

$$o_t = W_o \cdot (\sigma(r_t) \odot wkv_t), \quad (6)$$

where $\sigma(\cdot)$ represents sigmoid function and r_t is the receptance vector at t time stamp.

Similar to the Time Mixing sub-block, the Channel Mixing sub-block mixes different channel features by a non-linear mapping. k' and r' are calculated similarly to Eq. 4 but the output o' is different.

$$o'_t = \sigma(r'_t) \odot (W'_o \cdot \max(k'_t, 0)^2), \quad (7)$$

where we adopt the squared ReLU activation function[27].

Finally, the stock local embedding $h_{u,t}, t \in [1, T]$ preserves both the current details and encodes the signals indicative of other time steps, which enables the modelling of inter-temporal correlations for the same stock.

3.3.2 Stock Correlation. After the above process, the inter-temporal correlations within each stock are not blurred, as we keep the series of temporal representations without performing temporal alignment operations such as summation. Based on this, we further model the inter-temporal correlations across stocks at the same time step [16, 41]. In contrast to previous work that used predefined relationships, we use a self-attention mechanism to model dynamic relationships between stocks. Specifically, we aggregate the embedding of stocks at the same time step $H_t = \|_{u \in S} h_{u,t}$, and perform self-attention operations along the stock dimension. The operations are defined as follows:

$$\begin{aligned} Q_t &= W_Q H_t, \quad K_t = W_K H_t, \quad V_t = W_V H_t, \\ Z_t &= \|_{u \in S} z_{u,t} = \text{FFN}(\text{MHA}(Q_t, K_t, V_t) + H_t). \end{aligned} \quad (8)$$

where $\|$ denotes the concatenation of vectors, W_Q, W_K and W_V are projection matrixes. MHA represents multi-head attention mechanisms. FFN denotes feed forward network, which is constructed by a two-layer MLP with ReLU activation and residual connection.

Eventually, we achieve the modelling of the cross-time correlation of different stocks at the same time step by performing self-attention operations on the stock axis. Each stock is allowed to exchange information with other stocks, autonomously establishing or dismantling correlations.

3.4 Stock Embedding Aggregation Module

In this module, we use the temporal attention layer like [16] to aggregate the temporal features of each stock to generate the final embedding of the stock e_u . This aggregation step ensures that the most relevant temporal information is captured and consolidated. For each stock's embedding $z_{u,t}, t \in [1, T]$, we use the last day's embedding $z_{u,T}$ as a query vector to compute attention scores with other temporal features within the window, and then aggregate the results as follows:

$$\lambda_{u,t} = \frac{\exp(z_{u,t}^T W_\lambda z_{u,T})}{\sum_{i=1}^T \exp(z_{u,i}^T W_\lambda z_{u,T})}, \quad e_u = \sum_{t \in [1, T]} \lambda_{u,t} z_{u,t}, \quad (9)$$

¹More evidence and details can be referred to RwkV: Reinventing rnnns for the transformer era.

where W_λ denotes the transformation matrix and $\lambda_{u,t}$ represents the attention score.

3.5 Prediction

Finally, the embedding of the stock e_u is fed into the predictor $g(\cdot)$ to output the prediction results. We use a linear layer as the predictor and evaluate the results by Mean Squared Error (MSE) loss function. In each batch, the model is jointly optimised for all $u \in S$ on a particular prediction date. A training epoch consists of multiple batches that correspond to different prediction dates in the training set.

$$\hat{r}_u = g(e_u), \quad Loss = \sum_{u \in S} MSE(r_u, \hat{r}_u). \quad (10)$$

4 Experiment

In this section, we conduct experiments to answer the following four research questions:

- **RQ1:** How is the overall performance of MATCC compared with state-of-the-art methods?
- **RQ2:** Is the proposed stock trend and inter-temporal correlation modules effective for stock price forecasting?
- **RQ3:** How do hyper-parameter configurations and market noise affect the performance of MATCC?
- **RQ4:** What insights on the stock trend and cross-time correlation can we get through visualizing the attention map?

4.1 Experiment Settings

4.1.1 Dataset. We evaluate our framework on three real-world datasets from the US stock market and Chinese stock market. CSI300 and CSI800 represent two sets of stocks comprising the top 300 and 800 stocks, respectively, with the highest capital value on the Shanghai Stock Exchange and the Shenzhen Stock Exchange. The S&P500 dataset contains a broad range of publicly traded U.S. companies with more diversified risks that reflect broader market changes. All datasets encompass daily information from 2008 to 2023. We use the data from Q1 2008 to Q1 2020 as the training set, data in Q2 2020 as the validation set, and Q3 2020 to Q4 2023, are reserved as the test set. We apply the public Alpha158 indicators [40] to extract stock features from the collected data. Following [16], the lookback window length T and prediction interval d are set as 8 and 5 respectively. For Chinese market representation, we construct $F' = 63$ features with CSI300, CSI500 and CSI800 market indices while we include GSPC, DJI and NDX market indices for the US market. The refereable interval $d' = 5, 10, 20, 30, 60$. We summarize the statistics of three datasets in Table 1.

Table 1: Dataset details.

Market	CSI300	CSI800	S&P500
Stocks	300	800	500
Training period	2008/01/01-2020/03/31		
Validation period	2020/04/01-2020/06/30		
Test period	2020/07/01-2023/12/31		
# Features	158 + 63		

4.1.2 Data preprocessing. In this section, we introduce some details of data preprocessing [10] for easier reproduction.

1) *Robust Z-score Normalization:* To address the issue of varying scales between factors, we adopt Robust Z-score Normalization within stocks. This approach replaces the mean and standard deviation used in traditional z-score normalization with the median (MED) and the median absolute deviation (MAD). In robust statistical methods, the median is a robust measure of central tendency compared to the mean, while the median absolute deviation is a robust measure of statistical dispersion compared to the standard deviation. Specifically, the u th input stock data is normalized to:

$$\tilde{x}_u = \frac{|x_u - \text{MED}(X)|}{\text{MAD}(X)}. \quad (11)$$

2) *Dropping Extreme Label:* When a stock reaches its limit up (or limit down), stockholders tend to be more hesitant to sell (or buy) because they anticipate the trend to persist. Consequently, it becomes challenging for other stockholders to execute buy (or sell) orders. Therefore, it is impractical for the model to learn a strategy of "buying when there is a limit up and selling when there is a limit down." To address this issue, we suggest dropping extreme labels to mitigate the impact of these extreme values. Here we drop the first 2.5% and the last 2.5% percent labels.

4.1.3 Baselines. We compare the performance of MATCC with several stock price forecasting baselines from different categories.

- **LSTM** [11]: a Long Short-Term Memory network based stock price forecasting method.
- **ALSTM** [23]: an LSTM variant that incorporates a temporal attentive aggregation layer to aggregate information from hidden embeddings in previous timestamps.
- **GRU** [3]: a Gated Recurrent Unit (GRU) network based stock price forecasting method
- **GAT** [29]: a graph-based baseline, which uses sequential encoder to gain stock presentation and then aggregates information by graph attention networks.
- **Transformer** [28]: a transformer based stock price forecasting model.
- **DTML** [41]: a state-of-the-art stock correlation mining method, which adopts the attention mechanism to mine the dynamic correlation among stocks and incorporates the market information into the modeling.
- **MASTER** [16]: a state-of-the-art stock price forecasting model, which uses the transformer to model the momentary and cross-time stock correlation and leverages market information for automatic feature selection.

4.1.4 Evaluation. In our evaluation of model performance, we employ both ranking metrics and portfolio-based metrics to provide a comprehensive assessment. Four ranking metrics are considered: Information Coefficient (IC) [18], Rank Information Coefficient (RankIC) [17], Information Ratio based IC (ICIR), and Information Ratio based RankIC (RankICIR) [5]. IC denotes the Pearson correlation between prediction and label, while RankIC denotes the Spearman's rank correlation between prediction and label. ICIR and RankICIR are normalized metrics derived from IC and RankIC by dividing by the standard deviation. These metrics offer insights into

Table 2: Overall performance comparison on CSI300, CSI800 and S&P500 stock datasets. The best, second and third results are shown in red, blue and green.

Dataset	Model	IC $\uparrow$	ICIR $\uparrow$	RankIC $\uparrow$	RankICIR $\uparrow$	AR $\uparrow$	IR $\uparrow$
CSI300	LSTM	0.048 $\pm$ 0.008	0.34 $\pm$ 0.06	0.050 $\pm$ 0.010	0.34 $\pm$ 0.05	0.13 $\pm$ 0.04	1.3 $\pm$ 0.5
	GRU	0.046 $\pm$ 0.002	0.32 $\pm$ 0.02	0.046 $\pm$ 0.004	0.33 $\pm$ 0.02	0.11 $\pm$ 0.04	1.0 $\pm$ 0.4
	ALSTM	0.043 $\pm$ 0.004	0.30 $\pm$ 0.03	0.041 $\pm$ 0.004	0.30 $\pm$ 0.03	0.11 $\pm$ 0.03	1.1 $\pm$ 0.3
	Transformer	0.048 $\pm$ 0.002	0.32 $\pm$ 0.02	0.046 $\pm$ 0.003	0.32 $\pm$ 0.02	0.11 $\pm$ 0.02	1.0 $\pm$ 0.2
	GAT	0.052 $\pm$ 0.001	0.38 $\pm$ 0.02	0.051 $\pm$ 0.002	0.38 $\pm$ 0.02	0.17 $\pm$ 0.03	1.6 $\pm$ 0.2
	DTML	0.051 $\pm$ 0.004	0.35 $\pm$ 0.03	0.052 $\pm$ 0.006	0.35 $\pm$ 0.03	0.15 $\pm$ 0.02	1.5 $\pm$ 0.1
	MASTER	0.053 $\pm$ 0.004	0.39 $\pm$ 0.04	0.054 $\pm$ 0.006	0.39 $\pm$ 0.05	0.19 $\pm$ 0.02	1.9 $\pm$ 0.1
	Ours	0.117 $\pm$ 0.007	1.02 $\pm$ 0.06	0.086 $\pm$ 0.004	0.87 $\pm$ 0.06	0.80 $\pm$ 0.08	8.5 $\pm$ 0.8
CSI800	LSTM	0.043 $\pm$ 0.005	0.36 $\pm$ 0.05	0.043 $\pm$ 0.008	0.36 $\pm$ 0.05	0.15 $\pm$ 0.04	1.2 $\pm$ 0.4
	GRU	0.040 $\pm$ 0.002	0.35 $\pm$ 0.01	0.043 $\pm$ 0.002	0.35 $\pm$ 0.01	0.07 $\pm$ 0.02	0.6 $\pm$ 0.2
	ALSTM	0.039 $\pm$ 0.003	0.34 $\pm$ 0.02	0.041 $\pm$ 0.004	0.33 $\pm$ 0.02	0.11 $\pm$ 0.03	0.9 $\pm$ 0.3
	Transformer	0.038 $\pm$ 0.003	0.32 $\pm$ 0.03	0.040 $\pm$ 0.006	0.31 $\pm$ 0.04	0.10 $\pm$ 0.02	0.8 $\pm$ 0.2
	GAT	0.044 $\pm$ 0.004	0.39 $\pm$ 0.04	0.047 $\pm$ 0.006	0.38 $\pm$ 0.04	0.15 $\pm$ 0.03	1.2 $\pm$ 0.2
	DTML	0.043 $\pm$ 0.002	0.35 $\pm$ 0.02	0.047 $\pm$ 0.004	0.34 $\pm$ 0.02	0.14 $\pm$ 0.01	1.2 $\pm$ 0.1
	MASTER	0.046 $\pm$ 0.004	0.42 $\pm$ 0.05	0.048 $\pm$ 0.007	0.41 $\pm$ 0.05	0.16 $\pm$ 0.04	1.3 $\pm$ 0.3
	Ours	0.118 $\pm$ 0.007	1.33 $\pm$ 0.07	0.083 $\pm$ 0.004	1.07 $\pm$ 0.06	1.62 $\pm$ 0.11	13.3 $\pm$ 0.8
S&P500	LSTM	0.006 $\pm$ 0.001	0.03 $\pm$ 0.00	0.005 $\pm$ 0.001	0.03 $\pm$ 0.00	0.08 $\pm$ 0.03	0.8 $\pm$ 0.2
	GRU	0.012 $\pm$ 0.002	0.07 $\pm$ 0.01	0.010 $\pm$ 0.002	0.07 $\pm$ 0.01	0.07 $\pm$ 0.01	0.6 $\pm$ 0.1
	ALSTM	0.011 $\pm$ 0.002	0.06 $\pm$ 0.01	0.010 $\pm$ 0.003	0.05 $\pm$ 0.01	0.07 $\pm$ 0.01	0.6 $\pm$ 0.1
	Transformer	0.015 $\pm$ 0.001	0.08 $\pm$ 0.02	0.014 $\pm$ 0.001	0.08 $\pm$ 0.02	0.08 $\pm$ 0.03	0.6 $\pm$ 0.1
	GAT	0.014 $\pm$ 0.005	0.07 $\pm$ 0.02	0.013 $\pm$ 0.004	0.07 $\pm$ 0.02	0.09 $\pm$ 0.02	0.7 $\pm$ 0.2
	DTML	0.012 $\pm$ 0.001	0.10 $\pm$ 0.01	0.011 $\pm$ 0.001	0.10 $\pm$ 0.01	0.05 $\pm$ 0.01	0.5 $\pm$ 0.1
	MASTER	0.021 $\pm$ 0.001	0.14 $\pm$ 0.01	0.019 $\pm$ 0.001	0.14 $\pm$ 0.01	0.09 $\pm$ 0.02	0.9 $\pm$ 0.1
	Ours	0.071 $\pm$ 0.001	0.81 $\pm$ 0.02	0.035 $\pm$ 0.001	0.32 $\pm$ 0.01	0.64 $\pm$ 0.02	6.7 $\pm$ 0.2

Table 3: Ablation study to validate the effectiveness of proposed modules on CSI. We remove each module separately while maintaining other blocks in MATCC. Transformer and Mamba represent replacing RWKV with Transformer or Mamba in the Time Correlation. The best, second and third results are shown in red, blue and green.

Dataset	Model	IC $\uparrow$	ICIR $\uparrow$	RankIC $\uparrow$	RankICIR $\uparrow$	AR $\uparrow$	IR $\uparrow$
CSI300	w/o MTG	0.114 $\pm$ 0.010	1.01 $\pm$ 0.10	0.085 $\pm$ 0.004	0.96 $\pm$ 0.10	0.79 $\pm$ 0.05	8.4 $\pm$ 0.6
	w/o STD	0.048 $\pm$ 0.003	0.37 $\pm$ 0.02	0.048 $\pm$ 0.003	0.37 $\pm$ 0.02	0.12 $\pm$ 0.02	1.3 $\pm$ 0.3
	w/o TC	0.055 $\pm$ 0.008	0.42 $\pm$ 0.08	0.052 $\pm$ 0.006	0.35 $\pm$ 0.18	0.19 $\pm$ 0.09	2.0 $\pm$ 1.0
	w/o SC	0.049 $\pm$ 0.005	0.41 $\pm$ 0.05	0.046 $\pm$ 0.004	0.41 $\pm$ 0.05	0.16 $\pm$ 0.05	1.9 $\pm$ 0.5
	Change order	0.054 $\pm$ 0.002	0.39 $\pm$ 0.04	0.053 $\pm$ 0.003	0.39 $\pm$ 0.04	0.14 $\pm$ 0.03	1.4 $\pm$ 0.3
	Transformer	0.083 $\pm$ 0.006	0.65 $\pm$ 0.05	0.071 $\pm$ 0.003	0.65 $\pm$ 0.04	0.52 $\pm$ 0.08	5.3 $\pm$ 0.8
	Mamba	0.052 $\pm$ 0.003	0.38 $\pm$ 0.03	0.051 $\pm$ 0.002	0.29 $\pm$ 0.16	0.13 $\pm$ 0.06	1.3 $\pm$ 0.7
	Ours	0.117 $\pm$ 0.007	1.02 $\pm$ 0.06	0.086 $\pm$ 0.004	0.87 $\pm$ 0.06	0.80 $\pm$ 0.08	8.5 $\pm$ 0.8
CSI800	w/o MTG	0.087 $\pm$ 0.032	0.91 $\pm$ 0.42	0.068 $\pm$ 0.017	0.83 $\pm$ 0.33	1.11 $\pm$ 0.72	9.1 $\pm$ 5.8
	w/o STD	0.043 $\pm$ 0.002	0.42 $\pm$ 0.02	0.047 $\pm$ 0.002	0.42 $\pm$ 0.02	0.14 $\pm$ 0.04	1.2 $\pm$ 0.4
	w/o TC	0.047 $\pm$ 0.003	0.43 $\pm$ 0.03	0.048 $\pm$ 0.004	0.42 $\pm$ 0.03	0.14 $\pm$ 0.06	1.2 $\pm$ 0.6
	w/o SC	0.061 $\pm$ 0.029	0.63 $\pm$ 0.33	0.051 $\pm$ 0.015	0.60 $\pm$ 0.29	0.49 $\pm$ 0.60	4.8 $\pm$ 5.8
	Change order	0.052 $\pm$ 0.009	0.46 $\pm$ 0.11	0.051 $\pm$ 0.006	0.45 $\pm$ 0.11	0.24 $\pm$ 0.11	2.3 $\pm$ 1.1
	Transformer	0.097 $\pm$ 0.015	0.98 $\pm$ 0.19	0.079 $\pm$ 0.007	0.91 $\pm$ 0.13	1.25 $\pm$ 0.38	10.4 $\pm$ 2.6
	Mamba	0.046 $\pm$ 0.002	0.42 $\pm$ 0.03	0.048 $\pm$ 0.001	0.42 $\pm$ 0.03	0.13 $\pm$ 0.03	1.2 $\pm$ 0.2
	Ours	0.118 $\pm$ 0.007	1.33 $\pm$ 0.07	0.083 $\pm$ 0.004	1.07 $\pm$ 0.06	1.62 $\pm$ 0.11	13.3 $\pm$ 0.8

the model’s ability to generate accurate rankings and provide normalized measures of performance [37, 40]. We calculate the average

score at the same timestamp. Furthermore, two portfolio-based metrics are employed to compare the investment profit and risk of each

method. We simulate the stock trade with "top30-drop30" strategy: "top30" means keeping the stocks with the top 30 return ratios; and "drop30" means dropping any stock whose score falls out of the top 30, regardless of its previous performance. We report the Excess Annualized Return (AR) and Information Ratio (IR) metrics without cost. AR measures the annual expected excess return generated by the investment, while IR measures the risk-adjusted performance of an investment.

4.1.5 Implementation. We implement MATCC with PyTorch and build our methods based on the open-source quantitative investment platform Qlib [40]. The preprocess of the data follows our baseline model, MASTER [16]. In particular, since the implementation of "DropExtremeLabel" is not given by [16], we implement it according to [10]. With regard to baselines, the optimal hyperparameter settings and learning rates are in accordance with those parameters provided by [16]. For hyper-parameters of MATCC, the model dimension D , DepthConv kernel size k_c , AvgPool kernel size k_a and WKV heads N are tuned from $\{128, 256, 512\}$, $\{3, 5, 7\}$, $\{3, 5, 7\}$ and $\{2, 4, 8\}$ respectively. We set $D = 256$, $k_c = 5$, $k_a = 3$ and $N = 4$ based on the IC performance in the validation stage. The model is trained using CosineAnnealingLR [20] with a warm-up stage and optimized by the Adam optimizer. The maximum and minimum learning rate are 3×10^{-3} and 2×10^{-4} respectively. The warm up epoch is set to 10, and the number of epochs between restarts is 15. A total of 70 epochs were trained, with the final epoch of parameters used for testing. All experiments were conducted on a server equipped with an Intel(R) Xeon(R) CPU E5-2678 v3 (48 CPUs, 125GB) and a NVIDIA GeForce RTX 3090 GPU (24GB). We run the training and testing procedure 5 times for all methods and report average values and standard deviations.

5 Results

5.1 Overall Performance (RQ1)

The overall performance is reported in Table 2. On three datasets from different stock markets, MATCC far outperforms the baseline models. We find that all models perform better in Chinese stocks than in the US stock market because the stock trading mechanism and market rules differ. Despite the complicated market environment, MATCC shows a 120% improvement in ranking metrics and a 300% improvement in portfolio-based metrics compared to MASTER (SOTA) in 2024. It is worth noting that MASTER and DTML, which also use market information and stock correlation modelling, fail to perform well, reinforcing the important role of market and stock trends in forecasting. We also note that the *std/mean* ratio for most metrics is in the 5% to 10% range, demonstrating that MATCC has good stability and reliability. We attribute this to the RWKV and Market Trend Guidance module, which the ablation experiments can verify.

5.2 Model Study (RQ2)

To demonstrate the effectiveness of each additional component in MATCC, we conduct an ablation study over different model variants on Chinese stock datasets due to limited space and resources. We remove one additional component for each variant, *i.e.*, no Market Trend Guidance (**w/o MTG**); no Stock Trend Decomposition

(**w/o STD**); no Time Correlation (**w/o TC**); no Stock Correlation (**w/o SC**). We also verify the effect of exchange time correlation and stock correlation on the model performance (**Change order**). To further validate the superiority of RWKV, we separately use the classical self-attention module Transformer with the same heads (**Transformer**) and an improved RNN attention structure model Mamba [12] (**Mamba**) to model temporal correlation instead of RWKV. The results in Table 3 illustrate the efficacy of our tailored stock trend modules and RWKV. From the table, we make the following observations:

- When the Market Trend Guidance is removed, the impact on the CSI300 is small but significant on the CSI800, especially on the average score and the *std/mean* ratio. The overall trend is more stable in the CSI300 as it includes the 300 most extensive stocks in terms of market capitalisation and trading volume; however, the CSI800 contains more small and medium-sized stocks, whose movements are more complex and volatile. As a result, the stability and performance of the forecast results will be compromised when the global market trend guidance is lost.
- When the Stock Trend Decomposition is excluded, the test performance on both datasets decreases significantly, demonstrating that stock trends can effectively improve the predictive power. This justifies our original intention to separate the stock data into trend and fluctuation components and aligns with the public's intuition about stock movements. It is worth noting that the *std/mean* ratio is still within 5%~10%, suggesting that market trends and RWKV can reduce the volatility of the results.
- When modelling inter-temporal correlations, the order of the time correlation and stock correlation significantly impacts the results, mainly because we explicitly use stock trend information. By modelling inter-temporal correlations within stocks first, we can maximise the use of trend information and avoid mutual interference between the trends of different stocks.
- When the Time Correlation is removed, the test results suffer a dramatic decrease, mainly because the model cannot model characteristics within stocks across time. When substituting the RWKV with the Transformer or Mamba to model time correlation, the average performance and stability were not as good as the results of our tailored MATCC, proving the RWKV module's superior performance.
- When the Stock Correlation is removed, the model loses information on the inter-temporal correlations between different stocks, and therefore, results drop significantly.

5.3 Sensitivity Analysis (RQ3)

5.3.1 Hyper-parameter sensitivity. To investigate the effect of different hyperparameter settings on MATCC, we perform a sensitivity study on DepthConv's kernel k_c , AvgPool's kernel k_a and RWKV's head N . As shown in Figure 3, when k_c is small, the module can only focus on a shorter period, and the market trend tends to change insignificantly; when k_c is large, it can lead to the loss or blurring of market information. For the pooling kernel k_a , larger kernels tend to lead to excessive smoothing. For the number of heads N of RWKV, an appropriate number of heads allows the model to fully use the features, while too large or too small heads can lead to a decline in effectiveness.

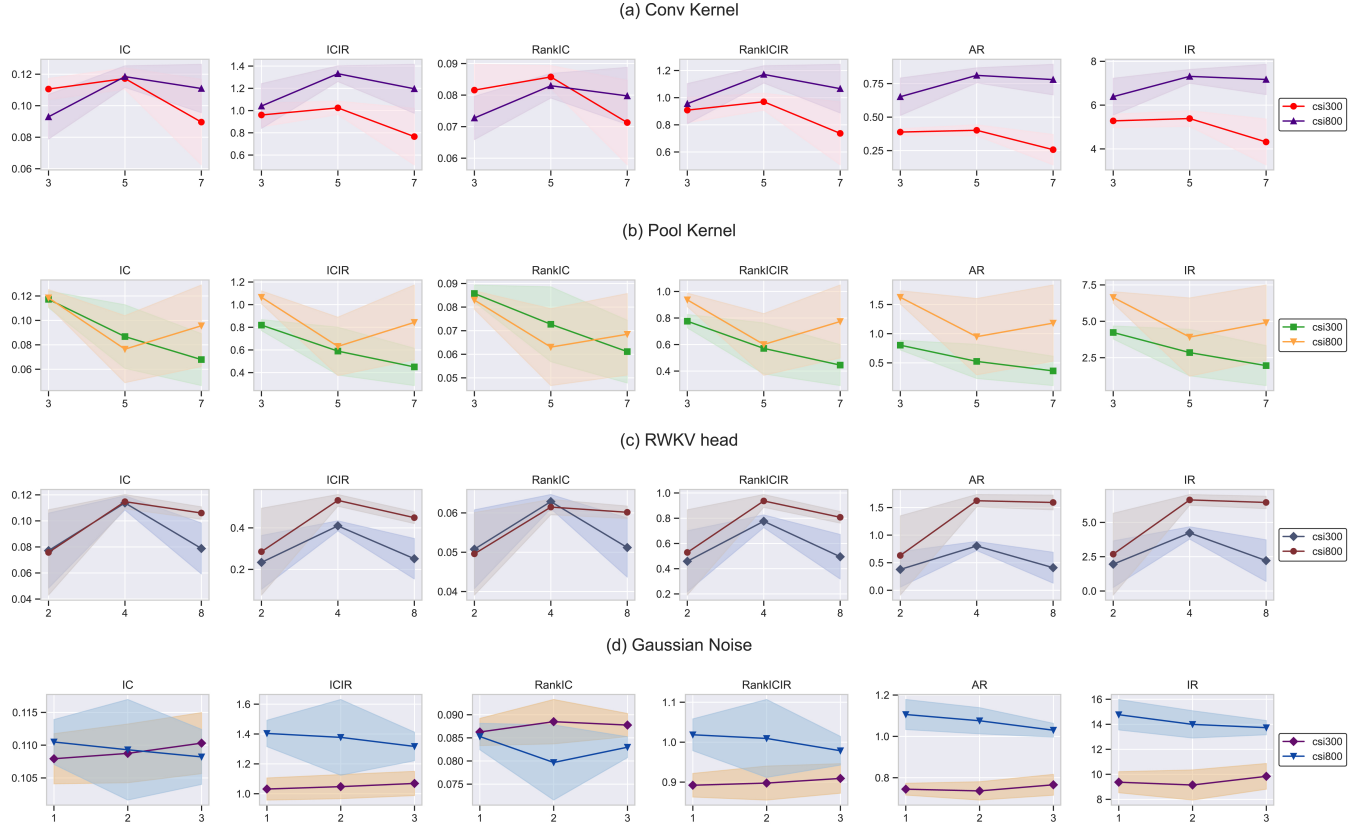
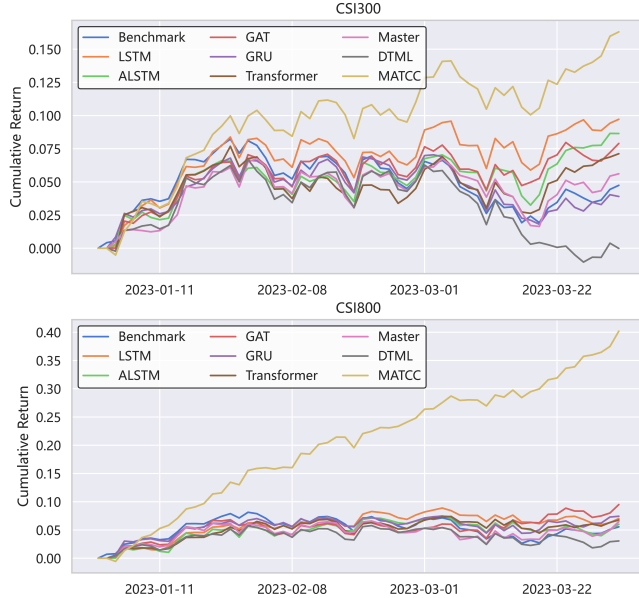
Figure 3: MATCC performance with varying k_c , k_a , N and Gaussian noise std γ .

Figure 4: Cumulative return comparison in a normal market environment.

5.3.2 Market noise sensitivity. Next, we examine the influence of market noise on MATCC to explore the model’s ability to handle extreme market conditions. We add the Gaussian random noise to the market features with $mean = 0$ and $std = \gamma$, $\gamma \in \{1.0, 2.0, 3.0\}$. γ increases from small to large, representing increasing market volatility. The results are shown in Figure 3 (d). Note that elements of the market representation result from a normalised process, and most of them are between $[-3, 3]$. We find that MATCC continues outperforming despite market noise and substantial volatility.

5.3.3 Portfolio Results. To fully validate the model’s profitability, we conduct market simulation trades under normal and extreme market conditions. We compare the cumulative profit curves corresponding to the different methods and the benchmark. Figure 4 shows that MATCC achieves the highest yield over almost the entire testing period despite the market volatility. As shown in Figure 5, under extreme market conditions, other models have all experienced significant negative returns during extreme downturns. In contrast, our model maintained positive returns for most of the down cycle especially on the CSI800, which is a strong indication of its robustness and profitability.

5.4 Visualization of Attention Maps (RQ4)

5.4.1 Time Correlation Attention Map. We will reveal how MATCC captures the long-term and short-term trends that previous

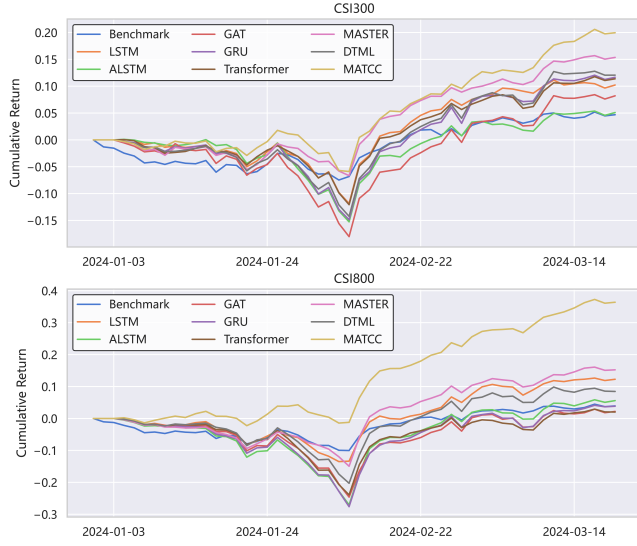


Figure 5: Cumulative return comparison in an extreme market environment.

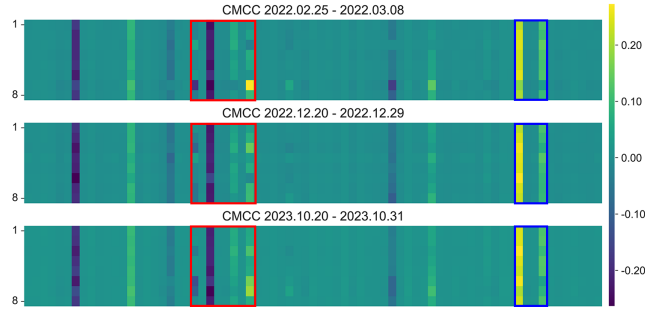


Figure 6: WKV attention map visualization. The x-axis is the stock features and the y-axis is the time steps in the lookback window.

methods are not expressive enough to model. Figure 6 shows the WKV attention map at different time steps in the lookback window. We chose the representative stock CMCC (SH600941) as the target and sampled 3 time periods for visualisation. The x-axis of the figure is stock features, and the y-axis is the time steps. Within the red line box, the highlighted portion diminishes and then intensifies over time, implying a change in trend pattern for this part of features as time advances. However, inside the blue box, these two indicators stay high, indicating that their trends remain the same as time passes. This map suggests that the Time Correlation module can capture relations within stocks across time.

5.4.2 Stock Correlation Attention Map. Figure 7 shows the attention map in the Stock Correlation part, learned by MATCC for the CSI800 dataset. Each element represents the amount of influence that stock j gives to stock i . Following [41], we reorder the stocks by the influence score $f(u) = \sum_i S_{iu}$ to compare the overall influence of stocks. The figure shows that several stocks have a

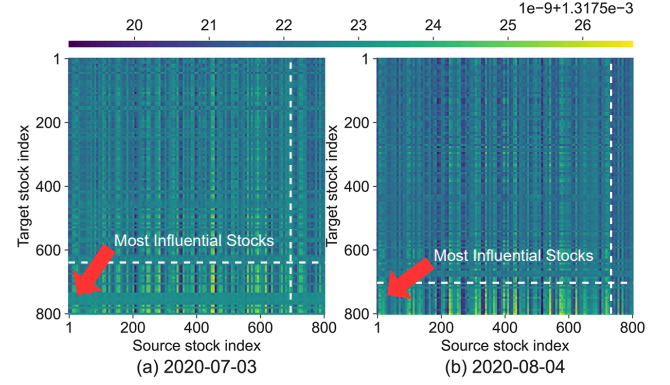


Figure 7: Stock correlation attention map. The stocks are ordered by the influence score $f(u) = \sum_i S_{iu}$. The most influential stocks are in the areas where the highlights are concentrated.

significant influence on the market, which supports our motivation that capturing dynamic cross-time correlations is essential for modelling accurate relationships between stocks.

6 Discussion

Due to diverse stock trading mechanisms and market rules in different countries, how to fully capture the market information requires more granular modelling of the unique trading features of each market. Considering space and resource constraints, we focus our experiments on real stock markets in China and the United States. More test results from other stock markets will be shown in our code repository.

7 Conclusion and Future Work

We propose a novel stock price prediction model MATCC, which fully uses market trend and stock trend information and model cross-time correlation. MATCC consists of six steps: market trend guidance, stock trend decomposition, time correlation, stock correlation, stock embedding aggregation and prediction. Experiments on the Chinese and US stock markets show that MATCC achieves average 100% improvements on ranking metrics and 300% on portfolio-based metrics compared with all baselines. At the same time, MATCC has managed to stabilise its performance in highly volatile market conditions. Visualisation of attention maps reveals trend changes and cross-time stock correlation. In conclusion, we provide a more granular perspective for studying stock trends and correlations while indicating a practical market information application. In future endeavours, we aim to investigate diverse strategies for reducing volatility in models and data, including exploring diffusion-based approaches.

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